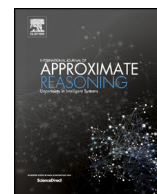




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Topological reduction approaches for relation decision systems

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ABSTRACT

In this study, we use topological methods to investigate attribute reduction problems for relation decision systems. We propose the notions of topological consistent and inconsistent relation decision systems. Then, we propose the concept of the consistent topological reduction of topological consistent relation decision systems. To enlarge the scope of application, we define the general topological reduction of relation decision systems, which is a further generalization of consistent topological reduction. Moreover, we develop corresponding reduction algorithms for these two types of reductions. To demonstrate that our algorithms cannot be unified by the general reduction algorithm, we discuss the relationship between the consistent topological reducts and the reducts identified by the general reduction algorithm. We conduct numerical experiments on 11 UCI datasets to verify our theoretical results. The experimental results demonstrate that the proposed reduction algorithms are effective and practicable.

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1. Introduction

The theory of rough sets proposed by Pawlak [1,2] in 1982 is regarded as a potent means of information processing. It can be used to process uncertain and incomplete information, and has been applied extensively in machine learning, data mining and pattern recognition. However, Pawlak rough sets are based on equivalence relations, which are too restrictive for practical applications. To address this problem, numerous generalized models of Pawlak rough sets have been proposed [3–12]. As a significant generalization of Pawlak rough sets, relation-based generalized rough sets, in which more general relations are substituted for equivalence relations, have attracted a great amount of research interest [13–19]. For instance, Liu and Zhu [13] used the solitary set to study the relationship between the lower and upper approximations of relation-based generalized rough sets. Zhu and Wang [14] found that the lower and upper approximation operations generated by different relations are not the same. Yao [15] defined the lower and upper approximation operators based on arbitrary binary relations. Moreover, he discussed the properties of several types of relation-based generalized rough sets.

As a critical and effective data dimension reduction technology, attribute reduction is one of the crucial application fields of rough sets. The essence of attribute reduction is to identify the minimal subsets of attributes that have the same properties as all the attributes. After attribute reduction, unimportant and redundant attributes are removed. Numerous authors have investigated the topic of attribute reduction [20–32]. Chen [33] defined part reduction in information systems. Liu and Feng [34] provided a knowledge granularity reduction algorithm that can identify all reduction sets. Moreover, they presented a simple algorithm for identifying a reduct. Fan et al. [35] proposed four types of reductions for multi-label data. At least two strategies exist for identifying attribute reduction sets. One of the strategies is the discernibility matrix-based approach. The concept of the discernibility matrix was proposed by Skowron and Rauszer [36]. They used discernibility matrices to identify all reducts for information systems. As an effective method, the discernibility matrix-based approach is very popular in the area of attribute reduction, and many authors have used this approach to compute attribute reduction sets [25,29,30,32–34,36–41].

Topology is a vital branch of mathematics that has a strong connection with rough sets. Using topological methods to study rough sets is an interesting topic, and is theoretically and practically important [42]. It has attracted much interest [43–50]. Pawlak [1,2] was the first researcher to investigate the relationships between topologies and rough sets. He demonstrated that an equivalence relation can induce a topology. With further research, many authors [47,51,52] proved that topologies can be induced by more general binary relations.

Relation decision systems, proposed by Wang et al. [53], are a significant extended form of decision tables [2,36]. However, in the initial definition, the decision attribute set of a relation decision system is composed of an equivalence relation, which is a constraint for practical applications. To address this issue, Liu et al. [54] provided a more general definition of relation decision systems in which the decision attribute set is composed of arbitrary binary relations. Moreover, they developed attribute reduction algorithms for consistent and inconsistent relation decision systems. However, the reduction algorithm for inconsistent relation decision systems may fail to identify all reducts. In [55], Liu et al. provided a general reduction algorithm (GRA) for relation decision systems to thoroughly solve this problem. Furthermore, the GRA unifies many of the previous reduction algorithms for usual decision tables.

To date, as we indicated in [32], attribute reduction problems in rough sets have not been studied intensively from topological viewpoints. The topologies on a universal set and the bases of a topology are two fundamental concepts in the theory of topology. Authors proposed various definitions of topological reduction in the literature [32,47,56,57], which can be divided into two categories: keeping a topology on a universal set unchanged and keeping one of the bases of a topology invariant. In [56], Lashin and Medhat defined the concept of topological reduction, which keeps one of the bases of a topology invariant, and offered a discernibility matrix to identify all reducts. However, Lin [58] showed that the reducts obtained by that discernibility matrix may not coincide with the definition of the reducts in [56]. Moreover, he defined a new discernibility matrix to solve this problem. The concept of topological reduction which keeps a topology on a universal set invariant was proposed by Yu and Zhan [47]. However, their definition of topological reduction is based on reflexive and symmetric relations, which limit theoretical and practical aspects. To address this issue, we provided a more general definition of topological reduction based on arbitrary binary relations in [32], which maintains a topology on a universal set invariant. Recently, Wu and Xu [57] proposed a new type of reduction called relative reduction. They presented topological reduction as a special case of relative reduction. Despite this, none of the knowledge representation systems in [32,47,56–58], including information systems, incomplete information systems, and relation systems, have decision attribute sets; that is, the topological reduction problems of decision systems in rough sets have not been investigated intensively. Specifically, topological reduction approaches for decision systems in rough sets have not been provided.

Because of the theoretical gap, we innovatively introduce topological reduction to decision systems in rough sets, and pioneer the study of topological reduction in relation decision systems. The introduction of the concept of topological reduction to decision systems is the main difference between the present study and [32]. To study the problems of topological reduction in relation decision systems, we first define topological consistent and inconsistent relation decision systems. Second, for a topological consistent relation decision system, we propose the concept of consistent topological reduction. Additionally, we propose a discernibility matrix associated with this type of reduction, and obtain the attribute reduction algorithm to identify all reducts. Third, to enlarge the applying scope, we define the concept of general topological reduction as a further generalization of consistent topological reduction. Then we develop the corresponding reduction algorithm based on a discernibility matrix to compute all reduction sets. Fourth, we study the relationship between the consistent

topological reducts and the reducts obtained by the GRA. We demonstrate that the GRA, which greatly unifies the previous attribute reduction algorithms for usual decision tables, cannot unify our reduction algorithms. Finally, we conduct comparative experiments on 11 UCI datasets to check our theoretical results.

The remainder of the paper is organized as follows. We review some basic concepts of rough sets, relation decision systems, and topologies, and provide the definitions of topological consistent and inconsistent relation decision systems in Section 2. In Section 3, we define the concept of consistent topological reduction and obtain the corresponding reduction algorithm. In Section 4, we define general topological reduction, and develop a correlative reduction algorithm. In Section 5, we discuss the relationship between the consistent topological reducts and the reducts identified by the GRA. In Section 6, we present numerical experiments to verify our theoretical results. We discuss our results in Section 7. Finally, Section 8 concludes the paper.

2. Preliminaries

In this section, we first recall some fundamental definitions of rough sets, relation decision systems, and topologies, and then define topological consistent and inconsistent relation decision systems.

Let $U = \{u_1, u_2, \dots, u_n\}$ be a non-empty finite set called the universal set, and R be a binary relation on U . Then the left and right R -relative sets of an element u in U are defined as

$$l_R(u) = \{v | v \in U, vRu\} \text{ and } r_R(u) = \{v | v \in U, uRv\},$$

respectively. We do not review the basic notions of reflexivity, symmetry, and transitivity of relations in this paper; we suggest that readers refer to [15,55]. With a binary relation R on U , for each $X \subseteq U$, Yao [15] defined the lower and upper approximations of X as

$$\underline{R}(X) = \{u | u \in U, r_R(u) \subseteq X\} \text{ and } \overline{R}(X) = \{u | u \in U, r_R(u) \cap X \neq \emptyset\},$$

respectively.

Definition 2.1. [54,55] Let U be a universal set, then $(U, C \cup D)$ is called a relation decision system, where $C = \{R_1, R_2, \dots, R_t\}$ and $D = \{d_1, d_2, \dots, d_k\}$ are two families of arbitrary binary relations on U , and $C \cap D = \emptyset$. C and D are called a condition attribute set and decision attribute set, respectively.

Hereafter, we assume that U is a non-empty finite universal set and $(U, C \cup D)$ is a relation decision system with $D = \{d\}$. The assumption $D = \{d\}$ does not influence our theoretical results. If $D = \{d_1, d_2, \dots, d_k\}$, then we can substitute $d = \bigcap_{d_i \in D} d_i$ for $d_1, d_2, \dots, d_k$.

For a universal set U , we use $P(U)$ to denote the power set of U .

Definition 2.2. [59] A subset τ of $P(U)$ is called a topology on U if it satisfies the following conditions:

- (1) $\emptyset, U \in \tau$.
 - (2) The intersection of any finite elements in τ is still an element in τ .
 - (3) The union of any arbitrary elements in τ is still an element in τ .
- Let τ be a topology on U . If $X \in \tau$ and $x \in X$, then X is called an open neighborhood of x .

Throughout this paper, for a binary relation S on U , we define a subset $\tau(S)$ of $P(U)$ as follows:

$$\tau(S) = \{X | X \subseteq U, X \subseteq \underline{S}(X)\}.$$

Proposition 2.1. [32] Let S be an arbitrary binary relation on U , then $\tau(S)$ is a topology on U .

The set $\tau(S)$ is called the topology induced by S . Let B be a set of binary relations on U . Then we use R_B to denote the intersection of all elements in B , that is, $R_B = \bigcap_{R \in B} R$. Clearly, R_B is a binary relation on U . By Proposition 2.1, for a relation decision system $(U, C \cup D)$, there are two topologies $\tau(R_C)$ and $\tau(R_D)$ on U , which are induced by binary relations R_C and R_D , respectively.

Definition 2.3. Let $(U, C \cup D)$ be a relation decision system. If $\tau(R_D) \subseteq \tau(R_C)$, then $(U, C \cup D)$ is called topological consistent; otherwise, $(U, C \cup D)$ is called topological inconsistent.

For a relation decision system $(U, C \cup D)$, we use $TCP_C(D)$ to denote the topological consistent part, which is defined as follows:

$$TCP_C(D) = \tau(R_C) \cap \tau(R_D).$$

It is easy to check that $TCP_C(D)$ is a topology on U .

3. Reduction approaches for topological consistent relation decision systems

In this section, we define the concept of consistent topological reduction. Additionally, we propose the corresponding reduction algorithm to compute all reducts.

We first provide a formal definition of consistent topological reduction.

Definition 3.1. Suppose that $(U, C \cup D)$ is topological consistent and $\emptyset \neq B \subseteq C$. Then B is called a consistent topological reduct of C if it satisfies the following conditions:

- (1) $\tau(R_D) \subseteq \tau(R_B)$.
- (2) If $\emptyset \neq A \subset B$, then $\tau(R_D) \not\subseteq \tau(R_A)$.

Clearly, condition (1) of Definition 3.1 is used to keep $(U, B \cup D)$ topological consistent, and condition (2) guarantees that B is a minimal subset of C that satisfies condition (1). Using condition (1) of Definition 3.1, $TCP_B(D) = \tau(R_B) \cap \tau(R_D) = \tau(R_D)$. Because $(U, C \cup D)$ is topological consistent, $TCP_C(D) = \tau(R_C) \cap \tau(R_D) = \tau(R_D)$. Hence, $TCP_C(D) = TCP_B(D)$. Therefore, the intuitive significance of consistent topological reduction is that it keeps the topology $TCP_C(D)$ invariant after reduction.

We require some lemmas to help us to prove Proposition 3.1 and Theorem 3.1.

Lemma 3.1. [52] $\cap_{i=1}^n \tau(S_i) = \tau(\cup_{i=1}^n S_i)$, where S_i is a binary relation on U .

We denote the transitive closure of a binary relation S on U by $t(S)$, and denote the relation of equality on U by I_U , that is, $I_U = \{(u, u) | u \in U\}$.

Lemma 3.2. [52] Suppose that S_1 and S_2 are binary relations on U . Then $\tau(S_1) = \tau(S_2)$ if and only if $t(S_1 \cup I_U) = t(S_2 \cup I_U)$.

Proposition 3.1. Suppose that $(U, C \cup D)$ is topological consistent. Then $t(R_C \cup R_D \cup I_U) = t(R_D \cup I_U)$.

Proof. Because $(U, C \cup D)$ is topological consistent, $\tau(R_D) \subseteq \tau(R_C)$. Hence, $\tau(R_C) \cap \tau(R_D) = \tau(R_D)$. Then, by Lemma 3.1, $\tau(R_C \cup R_D) = \tau(R_D)$. Thus, by Lemma 3.2, $t(R_C \cup R_D \cup I_U) = t(R_D \cup I_U)$. $\square$

Theorem 3.1. Suppose that $(U, C \cup D)$ is topological consistent and $\emptyset \neq B \subseteq C$. Then $\tau(R_D) \subseteq \tau(R_B)$ if and only if $t(R_B \cup R_D \cup I_U) = t(R_D \cup I_U)$.

Proof. Clearly, $\tau(R_D) \subseteq \tau(R_B)$ is equivalent to $\tau(R_B) \cap \tau(R_D) = \tau(R_D)$. Then, by Lemma 3.1, $\tau(R_D) = \tau(R_B) \cap \tau(R_D) = \tau(R_B \cup R_D)$. Hence, by Lemma 3.2, $\tau(R_B \cup R_D) = \tau(R_D)$ if and only if $t(R_B \cup R_D \cup I_U) = t(R_D \cup I_U)$. Thus, $\tau(R_D) \subseteq \tau(R_B)$ if and only if $t(R_B \cup R_D \cup I_U) = t(R_D \cup I_U)$. $\square$

Suppose that $(U, C \cup D)$ is topological consistent and $\emptyset \neq B \subseteq C$. Then, by Theorem 3.1, B is a consistent topological reduct of C if it satisfies the following conditions:

- (1) $t(R_B \cup R_D \cup I_U) = t(R_D \cup I_U)$.
- (2) If $\emptyset \neq A \subset B$, then $t(R_A \cup R_D \cup I_U) \neq t(R_D \cup I_U)$.

To develop a reduction algorithm, we define the discernibility matrix $M = (m_{ij})_{n \times n}$ for a topological consistent relation decision system $(U, C \cup D)$ as follows:

$$m_{ij} = \begin{cases} \{S | S \in C, (u_i, u_j) \notin S\}, & (u_i, u_j) \notin t(R_D \cup I_U) \\ \emptyset, & \text{otherwise} \end{cases},$$

where $u_i, u_j \in U$ and $n = |U|$. To obtain M , we first use the Warshall algorithm [60] to compute $t(R_D \cup I_U)$, for which the time complexity is $O(n^3)$. Then, we calculate each m_{ij} in M , for which the time complexity is $O(n^2)$. Hence, the time complexity of M is $O(n^3)$.

Remark. If R_D is a reflexive and transitive relation on U , then $M = (m_{ij})_{n \times n}$ can be rewritten as follows:

$$m_{ij} = \begin{cases} \{S | S \in C, (u_i, u_j) \notin S\}, & (u_i, u_j) \notin R_D \\ \emptyset, & \text{otherwise} \end{cases}.$$

Then the time complexity of M is $O(n^2)$.

We require a lemma to help us to prove Theorem 3.2.

Lemma 3.3. Suppose that $(U, C \cup D)$ is topological consistent. If $(u_i, u_j) \notin t(R_D \cup I_U)$, then $m_{ij} \neq \emptyset$.

Proof. Suppose that $m_{ij} = \emptyset$. Then for each $S \in C$, $(u_i, u_j) \in S$, which means that $(u_i, u_j) \in R_C$. Because $R_C \subseteq R_C \cup R_D \cup I_U \subseteq t(R_C \cup R_D \cup I_U)$, $(u_i, u_j) \in t(R_C \cup R_D \cup I_U)$. Because $(U, C \cup D)$ is topological consistent, $(u_i, u_j) \in t(R_C \cup R_D \cup I_U) = t(R_D \cup I_U)$ by Proposition 3.1. This contradicts $(u_i, u_j) \notin t(R_D \cup I_U)$. $\square$

Theorem 3.2. Suppose that $(U, C \cup D)$ is topological consistent and $\emptyset \neq B \subseteq C$. Then the following conditions are equivalent:

- (1) $t(R_B \cup R_D \cup I_U) = t(R_D \cup I_U)$.
- (2) If $m_{ij} \neq \emptyset$, then $m_{ij} \cap B \neq \emptyset$.

Proof. (1) $\Rightarrow$ (2): Given the definition of m_{ij} , if $m_{ij} \neq \emptyset$, then $(u_i, u_j) \notin t(R_D \cup I_U)$. Suppose that $m_{ij} \cap B = \emptyset$. Then $(u_i, u_j) \in R_B$. Hence, $(u_i, u_j) \in R_B \subseteq t(R_B \cup R_D \cup I_U)$. Then, using condition (1), $(u_i, u_j) \in t(R_D \cup I_U)$. This is a contradiction.

(2) $\Rightarrow$ (1): Because $R_D \cup I_U \subseteq R_B \cup R_D \cup I_U$, $t(R_D \cup I_U) \subseteq t(R_B \cup R_D \cup I_U)$. We will show that $t(R_B \cup R_D \cup I_U) \subseteq t(R_D \cup I_U)$. For arbitrary $(u_i, u_j) \notin t(R_D \cup I_U)$, $m_{ij} \neq \emptyset$ by Lemma 3.3. Then, using condition (2), $m_{ij} \cap B \neq \emptyset$; that is, $\exists S \in B$ such that $(u_i, u_j) \in S$; hence, $(u_i, u_j) \in R_B$. This implies that $R_B \subseteq t(R_D \cup I_U)$. Because $R_D \cup I_U \subseteq t(R_D \cup I_U)$ and $R_B \subseteq t(R_D \cup I_U)$, $R_B \cup R_D \cup I_U \subseteq t(R_D \cup I_U)$; hence, $t(R_B \cup R_D \cup I_U) \subseteq t(R_D \cup I_U)$. Thus, $t(R_B \cup R_D \cup I_U) = t(R_D \cup I_U)$. $\square$

Corollary 3.1. Suppose that $(U, C \cup D)$ is topological consistent and $\emptyset \neq B \subseteq C$. Then B is a consistent topological reduct of C if and only if it is a minimal subset that satisfies $m_{ij} \cap B \neq \emptyset$ for any $m_{ij} \neq \emptyset$.

By Corollary 3.1, we can develop a reduction algorithm for a topological consistent relation decision system $(U, C \cup D)$. Let

$$T(M) = \{m_{ij} | m_{ij} \neq \emptyset, m_{ij} \text{ be an entry in } M\}.$$

Clearly, $(T(M), \subseteq)$ is a poset with the set inclusion relation $\subseteq$. Let $T_{\min}(M)$ be the set of all minimal elements in $T(M)$. To simplify the computational procedure, we process all the elements in $T_{\min}(M)$ instead of all the elements in $T(M)$. The reduction algorithm is as follows:

Algorithm 3.1. The consistent topological reduction algorithm for topological consistent relation decision systems.

Input: A topological consistent relation decision system $(U, C \cup D)$.

Output: All consistent topological reducts of C .

- (1) Calculate the discernibility matrix $M = (m_{ij})_{n \times n}$.
 - (2) Compute the minimal element set $T_{\min}(M)$.
 - (3) Transform the discernibility function f from its conjunctive normal form (CNF) $f = \prod_{m_{ij} \in T_{\min}(M)} (\sum m_{ij})$ into a disjunctive normal form (DNF) $f = \sum_{q=1}^p (\prod B_q)$, $(B_q \subseteq C)$.
 - (4) Acquire all consistent topological reducts $B_1, B_2, \dots, B_p$.
- End the algorithm.

It is widely known that step (3) of Algorithm 3.1 is time consuming. However, in recent years, Borowik and Luba [61] proposed a fast algorithm for transforming the discernibility function from its CNF into a DNF, which can be used to improve the time complexity of Algorithm 3.1.

We use the following example to illustrate our reduction algorithm.

Example 3.1. Let $(U, C \cup D)$ be a topological consistent relation decision system, where $U = \{u_1, u_2, u_3, u_4, u_5\}$, $C = \{R_1, R_2, R_3, R_4, R_5\}$, and $D = \{d\}$. The relational matrices of R_1, R_2, R_3, R_4, R_5 , and R_D are, respectively, as follows:

$$M_{R_1} = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \end{pmatrix}, M_{R_2} = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{pmatrix}, M_{R_3} = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \end{pmatrix}, M_{R_4} = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}, M_{R_5} = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \end{pmatrix}, \text{ and } M_{R_D} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}. \text{ Because } R_C = \bigcap_{i=1}^5 R_i, \text{ we calculate the relational matrix of } R_C \text{ as}$$

$$M_{R_C} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}. \text{ It is easy to check that } \tau(R_D) \subseteq \tau(R_C). \text{ Hence, } (U, C \cup D) \text{ is topological consistent. By calculation,}$$

$M_{t(R_D \cup I_U)} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}$, where $M_{t(R_D \cup I_U)}$ is the relational matrix of $t(R_D \cup I_U)$. Then, we calculate the discernibility

matrix, which is listed as follows:

$$M = \begin{pmatrix} \emptyset & \{R_1, R_2, R_5\} & \{R_3, R_4\} & \{R_2, R_4\} & C \\ \emptyset & \emptyset & \emptyset & \emptyset & \emptyset \\ \emptyset & \{R_1, R_2, R_3, R_5\} & \emptyset & \{R_1, R_4, R_5\} & \emptyset \\ \emptyset & \emptyset & \emptyset & \emptyset & \emptyset \\ \emptyset & \{R_1, R_4, R_5\} & \{R_2, R_4\} & \{R_3, R_4\} & \emptyset \end{pmatrix}.$$

Hence, $T_{min}(M) = \{\{R_2, R_4\}, \{R_3, R_4\}, \{R_1, R_2, R_5\}, \{R_1, R_4, R_5\}\}$. Thus, CNF $f = (R_2 + R_4)(R_3 + R_4)(R_1 + R_2 + R_5)(R_1 + R_4 + R_5)$. We apply Boolean operations and obtain DNF $f = R_1R_4 + R_2R_4 + R_4R_5 + R_1R_2R_3 + R_2R_3R_5$. Thus, $\{R_1, R_4\}$, $\{R_2, R_4\}$, $\{R_4, R_5\}$, $\{R_1, R_2, R_3\}$, and $\{R_2, R_3, R_5\}$ are all consistent topological reducts of C .

4. Reduction approaches for general relation decision systems

For a topological consistent relation decision system $(U, C \cup D)$, the condition $\tau(R_D) \subseteq \tau(R_C)$ may be a restriction on many practical problems, which limits the application of consistent topological reduction. To address this problem, in this section, we define the general topological reduction of relation decision systems as a further generation of consistent topological reduction. Moreover, we develop the corresponding reduction algorithm to identify all general topological reducts. The formal definition of general topological reduction is given as follows:

Definition 4.1. Let $(U, C \cup D)$ be a relation decision system and $\emptyset \neq B \subseteq C$. Then B is called a general topological reduct of C if it satisfies the following conditions:

- (1) $TCP_B(D) = TCP_C(D)$.
- (2) If $\emptyset \neq A \subset B$, then $TCP_A(D) \neq TCP_C(D)$.

Clearly, condition (1) of Definition 4.1 is used to keep the topology $TCP_C(D)$ invariant after reduction, which is the intuitive significance of general topological reduction. Condition (2) of Definition 4.1 ensures that B is a minimal subset of C that meets condition (1).

Remark. The intuitive significance of general topological reduction is the same as that of consistent topological reduction. If $(U, C \cup D)$ is topological consistent, then condition (1) of Definition 4.1 is equivalent to condition (1) of Definition 3.1. Therefore, the general topological reduction is a natural generalization of the consistent topological reduction.

Theorem 4.1. Let $(U, C \cup D)$ be a relation decision system and $\emptyset \neq B \subseteq C$. Then $TCP_B(D) = TCP_C(D)$ if and only if $t(R_B \cup R_D \cup I_U) = t(R_C \cup R_D \cup I_U)$.

Proof. The proof is analogous to that of Theorem 3.1; hence, we omit it. $\square$

Let $(U, C \cup D)$ be a relation decision system and $\emptyset \neq B \subseteq C$. By Theorem 4.1, B is a general topological reduct of C if it satisfies the following conditions:

- (1) $t(R_B \cup R_D \cup I_U) = t(R_C \cup R_D \cup I_U)$.
- (2) If $\emptyset \neq A \subset B$, then $t(R_A \cup R_D \cup I_U) \neq t(R_C \cup R_D \cup I_U)$.

To obtain a reduction algorithm that can identify all general topological reducts, we define the discernibility matrix $M = (m_{ij})_{n \times n}$ for a relation decision system $(U, C \cup D)$ as follows:

$$m_{ij} = \begin{cases} \{S | S \in C, (u_i, u_j) \notin S\}, & (u_i, u_j) \notin t(R_C \cup R_D \cup I_U) \\ \emptyset, & \text{otherwise} \end{cases},$$

where $u_i, u_j \in U$ and $n = |U|$. Because $t(R_C \cup R_D \cup I_U)$ is calculated using the Warshall algorithm, the time complexity of M is $O(n^3)$.

Remark. If $R_C \cup R_D$ is a reflexive and transitive relation on U , then $M = (m_{ij})_{n \times n}$ can be simplified as follows:

$$m_{ij} = \begin{cases} \{S | S \in C, (u_i, u_j) \notin S\}, & (u_i, u_j) \notin R_C \cup R_D \\ \emptyset, & \text{otherwise} \end{cases},$$

and then the time complexity of M is $O(n^2)$.

Table 1
Decision table.

U	c_1	c_2	c_3	c_4	c_5	d
u_1	0	0	0	0	1	1
u_2	1	1	0	2	0	1
u_3	0	0	0	0	1	2
u_4	1	1	1	1	2	2
u_5	1	2	1	2	2	3

The next lemma will help us to prove Theorem 4.2.

Lemma 4.1. Let $(U, C \cup D)$ be a relation decision system. If $(u_i, u_j) \notin t(R_C \cup R_D \cup I_U)$, then $m_{ij} \neq \emptyset$.

Proof. The proof is analogous to that of Lemma 3.3, hence we omit it. $\square$

Theorem 4.2. Let $(U, C \cup D)$ be a relation decision system and $\emptyset \neq B \subseteq C$. Then the following conditions are equivalent:

- (1) $t(R_B \cup R_D \cup I_U) = t(R_C \cup R_D \cup I_U)$.
- (2) If $m_{ij} \neq \emptyset$, then $m_{ij} \cap B \neq \emptyset$.

Proof. The proof is analogous to that of Theorem 3.2; hence, we omit it. $\square$

Corollary 4.1. Let $(U, C \cup D)$ be a relation decision system and $\emptyset \neq B \subseteq C$. Then B is a general topological reduct of C if and only if it is a minimal subset satisfying $m_{ij} \cap B \neq \emptyset$ for any $m_{ij} \neq \emptyset$.

By Corollary 4.1, we propose a reduction algorithm to identify all general topological reducts for a relation decision system as follows:

Algorithm 4.1. The general topological reduction algorithm for relation decision systems.

Input: A relation decision system $(U, C \cup D)$.

Output: All general topological reducts of C .

(1) Calculate the discernibility matrix $M = (m_{ij})_{n \times n}$.

(2) Compute the minimal element set $T_{\min}(M)$.

(3) Transform the discernibility function f from its CNF $f = \prod_{m_{ij} \in T_{\min}(M)} (\sum m_{ij})$ into a DNF $f = \sum_{q=1}^p (\prod B_q)$, ($B_q \subseteq C$).

(4) Acquire all general topological reducts $B_1, B_2, \dots, B_p$.

End the algorithm.

Remark. It is clear that Algorithm 4.1 is a generalization of Algorithm 3.1.

Decision tables are one of the most common and critical relation decision systems. We use the following example to illustrate the general topological reduction algorithm using a decision table.

Example 4.1. Let $(U, C' \cup D')$ be the decision table represented in Table 1. Then each attribute $b \in C' \cup D'$ can induce an equivalence relation R_b on U , where $R_b = \{(u_i, u_j) | u_i, u_j \in U, b(u_i) = b(u_j)\}$. Hence, the decision table $(U, C' \cup D')$ can be considered as a relation decision system $(U, C \cup D)$, where $C = \{R_{c_1}, R_{c_2}, R_{c_3}, R_{c_4}, R_{c_5}\}$ and $D = \{R_d\}$. By calculation, the relational matrices of $R_{c_1}, R_{c_2}, R_{c_3}, R_{c_4}, R_{c_5}, R_C$, and R_D are, respectively, as follows:

$$M_{R_{c_1}} = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix}, M_{R_{c_2}} = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, M_{R_{c_3}} = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}, M_{R_{c_4}} = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}, M_{R_{c_5}} = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}, M_{R_C} = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \text{ and } M_{R_D} = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \text{ Let } X = \{u_1, u_2\}. \text{ Then } X \notin \tau(R_C) \text{ and } X \in \tau(R_D). \text{ Thus, } \tau(R_D) \not\subseteq \tau(R_C), \text{ that is, } (U, C \cup D) \text{ is a topological inconsistent relation decision system. Then we compute}$$

the relational matrix of $t(R_C \cup R_D \cup I_U)$ as $M_{t(R_C \cup R_D \cup I_U)} = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$. Then we compute and list the discernibility

matrix for the general topological reduction as follows:

$$M = \begin{pmatrix} \emptyset & \emptyset & \emptyset & \emptyset & C \\ \emptyset & \emptyset & \emptyset & \emptyset & \{R_{C_2}, R_{C_3}, R_{C_5}\} \\ \emptyset & \emptyset & \emptyset & \emptyset & C \\ \emptyset & \emptyset & \emptyset & \emptyset & \{R_{C_2}, R_{C_4}\} \\ C & \{R_{C_2}, R_{C_3}, R_{C_5}\} & C & \{R_{C_2}, R_{C_4}\} & \emptyset \end{pmatrix}.$$

Hence, $T_{\min}(M) = \{\{R_{C_2}, R_{C_4}\}, \{R_{C_2}, R_{C_3}, R_{C_5}\}\}$. Then CNF $f = (R_{C_2} + R_{C_4})(R_{C_2} + R_{C_3} + R_{C_5})$. We apply Boolean operations and obtain DNF $f = R_{C_2} + R_{C_3}R_{C_4} + R_{C_4}R_{C_5}$. Thus, $\{\{c_2\}, \{c_3, c_4\}, \{c_4, c_5\}\}$ is the set of all general topological reducts of C' in $(U, C' \cup D')$.

5. Relationship between the consistent topological reducts and the reducts identified by GRA

In this section, we study the relationship between the consistent topological reducts and the reducts identified by the GRA [55], which greatly unifies the previous attribute reduction algorithms for usual decision tables.

Definition 5.1. [54] Let $(U, C \cup D)$ be a relation decision system. If $R_C \subseteq R_D$, then $(U, C \cup D)$ is called a consistent relation decision system; otherwise, $(U, C \cup D)$ is called an inconsistent relation decision system.

Let $(U, C \cup D)$ be a relation decision system. Liu et al. [55] defined the consistent part $CP_C(D)$ of $(U, C \cup D)$ as follows:

$$CP_C(D) = \{u | r_{R_C}(u) \subseteq r_{R_D}(u), u \in U\}.$$

Moreover, they used the consistent part $CP_C(D)$ as the invariant to define the reduction.

Definition 5.2. [55] Let $(U, C \cup D)$ be a relation decision system and $\emptyset \neq B \subseteq C$. Then B is a reduct of C if it satisfies the following conditions:

- (1) $CP_B(D) = CP_C(D)$.
- (2) If $\emptyset \neq A \subset B$, then $CP_A(D) \neq CP_C(D)$.

The reduct B is a minimal subset of C that keeps the consistent part of $(U, C \cup D)$ unchanged.

We require some lemmas to help us to prove the following theorems, which reflect the relationship between the consistent topological reducts and the reducts identified by the GRA.

Lemma 5.1. Let $(U, C \cup D)$ be a consistent relation decision system and $\emptyset \neq B \subseteq C$. Then $CP_B(D) = CP_C(D)$ if and only if $R_B \subseteq R_D$.

Proof. Because $(U, C \cup D)$ is consistent, $R_C \subseteq R_D$. Thus, $CP_C(D) = U$. Suppose that $CP_B(D) = CP_C(D)$. Then $CP_B(D) = U$. Hence, given the definition of $CP_B(D)$, $r_{R_B}(u) \subseteq r_{R_D}(u)$ for each $u \in U$. This implies that $R_B \subseteq R_D$. Conversely, suppose that $R_B \subseteq R_D$. Then $r_{R_B}(u) \subseteq r_{R_D}(u)$ for each $u \in U$. Thus, $CP_B(D) = U = CP_C(D)$. $\square$

Lemma 5.2. [52] Let S_1 and S_2 be binary relations on U . If $S_1 \subseteq S_2$, then $\tau(S_2) \subseteq \tau(S_1)$.

Remark. If $(U, C \cup D)$ is a consistent relation decision system, then it is also a topological consistent relation decision system, but the inverse is not true.

Lemma 5.3. [52] Let $\tau(S)$ be the topology on U induced by S . Then for each $u \in U$, $r_{t(S \cup I_U)}(u)$ is the smallest open neighborhood of u .

Proposition 5.1. Let $(U, C \cup D)$ be a relation decision system and $\emptyset \neq B \subseteq C$. If R_D is a reflexive and transitive relation on U , then the following conditions are equivalent:

- (1) $t(R_B \cup R_D \cup I_U) = t(R_D \cup I_U)$.
- (2) $R_B \subseteq R_D$.

Proof. Because R_D is a reflexive and transitive relation on U , $t(R_D \cup I_U) = t(R_D) = R_D$ and $t(R_B \cup R_D \cup I_U) = t(R_B \cup R_D)$.

(1) $\Rightarrow$ (2): Using $R_D = t(R_D \cup I_U) = t(R_B \cup R_D \cup I_U)$ and $t(R_B \cup R_D \cup I_U) = t(R_B \cup R_D)$, we obtain $t(R_B \cup R_D) = R_D$. Because $R_D \subseteq R_B \cup R_D \subseteq t(R_B \cup R_D) = R_D$, $R_B \cup R_D = R_D$, which implies that $R_B \subseteq R_D$.

(2) $\Rightarrow$ (1): Because $R_B \subseteq R_D$, $R_B \cup R_D \cup I_U = R_D \cup I_U$; hence, $t(R_B \cup R_D \cup I_U) = t(R_D \cup I_U)$. $\square$

Proposition 5.2. Let $(U, C \cup D)$ be a relation decision system. If R_C and R_D are reflexive and transitive relations on U , then $R_C \subseteq R_D$ if and only if $\tau(R_D) \subseteq \tau(R_C)$.

Proof. If $R_C \subseteq R_D$, then by Lemma 5.2, $\tau(R_D) \subseteq \tau(R_C)$. Conversely, because R_C and R_D are reflexive and transitive, $t(R_C \cup I_U) = R_C$ and $t(R_D \cup I_U) = R_D$. Thus, for each $u \in U$, $r_{t(R_C \cup I_U)}(u) = r_{R_C}(u)$ and $r_{t(R_D \cup I_U)}(u) = r_{R_D}(u)$. Then, by Lemma 5.3, for each $u \in U$, $r_{R_C}(u)$ is the smallest open neighborhood of u in $\tau(R_C)$ and $r_{R_D}(u)$ is the smallest open neighborhood of u in $\tau(R_D)$. Using $\tau(R_D) \subseteq \tau(R_C)$, given $r_{R_D}(u) \in \tau(R_D)$, $r_{R_D}(u) \in \tau(R_C)$ for each $u \in U$. Because $r_{R_C}(u)$ is the smallest open neighborhood of u in $\tau(R_C)$, $r_{R_C}(u) \subseteq r_{R_D}(u)$ for each $u \in U$. Thus, $R_C \subseteq R_D$. $\square$

Theorem 5.1. Let $(U, C \cup D)$ be a consistent relation decision system and $\emptyset \neq B \subseteq C$. If R_D is a reflexive and transitive relation on U , then $CP_B(D) = CP_C(D)$ if and only if $\tau(R_D) \subseteq \tau(R_B)$.

Proof. Because $(U, C \cup D)$ is consistent, $R_C \subseteq R_D$. Hence, $\tau(R_D) \subseteq \tau(R_C)$ by Lemma 5.2, that is, $(U, C \cup D)$ is also topological consistent. Given the consistency of $(U, C \cup D)$, $CP_B(D) = CP_C(D)$ if and only if $R_B \subseteq R_D$ by Lemma 5.1. Given that R_D is reflexive and transitive, by Proposition 5.1, $R_B \subseteq R_D$ if and only if $t(R_B \cup R_D \cup I_U) = t(R_D \cup I_U)$. Because $(U, C \cup D)$ is topological consistent, $\tau(R_D) \subseteq \tau(R_B)$ if and only if $t(R_B \cup R_D \cup I_U) = t(R_D \cup I_U)$ by Theorem 3.1. Thus, $CP_B(D) = CP_C(D)$ if and only if $\tau(R_D) \subseteq \tau(R_B)$. $\square$

Theorem 5.2. Suppose that $(U, C \cup D)$ is topological consistent and $\emptyset \neq B \subseteq C$. If R_C and R_D are reflexive and transitive relations on U , then $CP_B(D) = CP_C(D)$ if and only if $\tau(R_D) \subseteq \tau(R_B)$.

Proof. By Proposition 5.2, $(U, C \cup D)$ is a consistent relation decision system. Hence, by Theorem 5.1, $CP_B(D) = CP_C(D)$ if and only if $\tau(R_D) \subseteq \tau(R_B)$. $\square$

Theorems 5.1 and 5.2 illustrate the conditions under which the consistent topological reducts coincide with the reducts identified by the GRA. However, the relationship between the general topological reducts (not consistent topological reducts) and the reducts identified by the GRA remains an open problem.

The following example is used to demonstrate that the consistent topological reducts may not be the same as the reducts identified by the GRA if the conditions listed in Theorem 5.1 or 5.2 are not satisfied.

Example 5.1. Let $(U, C \cup D)$ be a consistent relation decision system, where $U = \{u_1, u_2, u_3, u_4, u_5\}$, $C = \{R_1, R_2, R_3, R_4, R_5\}$, and $D = \{d\}$. The relational matrices of R_1, R_2, R_3, R_4, R_5 , and R_D are, respectively, as follows:

$$M_{R_1} = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 \end{pmatrix}, M_{R_2} = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{pmatrix}, M_{R_3} = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \end{pmatrix}, M_{R_4} = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \end{pmatrix}, M_{R_5} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix},$$

$$\text{and } M_{R_D} = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}. \text{ Because } R_C = \bigcap_{i=1}^5 R_i, \text{ the relational matrix of } R_C \text{ is } M_{R_C} = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

Because R_C is not reflexive. Because $(u_1, u_4) \in R_D$ and $(u_4, u_2) \in R_D$, but $(u_1, u_2) \notin R_D$, R_D is not transitive. Thus, $(U, C \cup D)$ fails to satisfy the conditions listed in Theorem 5.1 or 5.2. By calculation, the relational matrix of $t(R_D \cup I_U)$ is

$$M_{t(R_D \cup I_U)} = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 \end{pmatrix}. \text{ Then we compute and list the discernibility matrix for the consistent topological reduction}$$

as follows:

$$M = \begin{pmatrix} \emptyset & \emptyset & \{R_2, R_3, R_4\} & \emptyset & \{R_1, R_2\} \\ \emptyset & \emptyset & \{R_2, R_3\} & \emptyset & \{R_2, R_3\} \\ C & \{R_1, R_2, R_5\} & \emptyset & C & C \\ \emptyset & \emptyset & C & \emptyset & \{R_2\} \\ \emptyset & \emptyset & \{R_1, R_3, R_5\} & \emptyset & \emptyset \end{pmatrix}$$

Table 2
Brief description of the 11 UCI datasets.

Serial numbers	Datasets	Samples	Condition attributes	Classes
1	Climate Model Simulation Crashes	540	18	2
2	Wine	178	13	3
3	Glass Identification	214	9	6
4	Leaf	340	14	30
5	Lymphography	148	18	4
6	Parkinsons	195	22	2
7	Speaker Accent Recognition	329	12	6
8	Turkish Music Emotion Dataset	400	50	4
9	SPECTF Heart(testing)	187	44	2
10	Planning Relax	182	12	2
11	Urban Land Cover(testing)	507	147	9

Then, $T_{min}(M) = \{\{R_2\}, \{R_1, R_3, R_5\}\}$. Hence, CNF $f = R_2(R_1 + R_3 + R_5)$. We apply Boolean operations and obtain DNF $f = R_1R_2 + R_2R_3 + R_2R_5$. Thus, $\{R_1, R_2\}$, $\{R_2, R_3\}$, and $\{R_2, R_5\}$ are all consistent topological reducts of C . Using the GRA, we obtain and list the discernibility matrix as follows:

$$M' = \begin{pmatrix} \emptyset & \{R_4\} & \{R_2, R_3, R_4\} & \emptyset & \{R_1, R_2\} \\ \emptyset & \emptyset & \{R_2, R_3\} & \emptyset & \{R_2, R_3\} \\ C & \{R_1, R_2, R_5\} & \emptyset & C & C \\ \{R_1, R_2\} & \emptyset & C & \emptyset & \{R_2\} \\ \{R_4, R_5\} & \emptyset & \{R_1, R_3, R_5\} & C & \emptyset \end{pmatrix}.$$

Then, $T_{min}(M') = \{\{R_2\}, \{R_4\}, \{R_1, R_3, R_5\}\}$. Hence, CNF $f' = R_2R_4(R_1 + R_3 + R_5)$. We apply Boolean operations and obtain DNF $f' = R_1R_2R_4 + R_2R_3R_4 + R_2R_4R_5$. Thus, $\{R_1, R_2, R_4\}$, $\{R_2, R_3, R_4\}$, and $\{R_2, R_4, R_5\}$ are all reducts of C . Clearly, the consistent topological reducts are different from the reducts identified by the GRA.

Remark. Example 5.1 shows that the GRA cannot unify Algorithm 3.1. Consequently, as a generalization of Algorithm 3.1, Algorithm 4.1 certainly cannot be unified by the GRA.

6. Experiments

To verify the effectiveness and practicability of the reduction algorithms, we conducted comparison experiments on 11 datasets selected from the UCI machine learning repository [62]. Some attributes were meaningless in the experiments, including “Latin hypercube study ID,” “Simulation ID,” “ID number,” “Specimen number,” and “Name”. Therefore, we deleted them from the correlative datasets. Information about the datasets involved in the experiments is summarized in Table 2.

The experimental datasets’ attributes comprised continuous attributes and discrete attributes (Boolean attributes or categorical attributes). We used the approach proposed in [32] to induce binary relations for the continuous attributes. Regarding the discrete attribute p , we used the following steps to induce a binary relation R_p :

- (1) Let T be a dataset, and u_i and u_j be instances of T .
- (2) If p is a Boolean attribute in T , then R_p can be induced via

$$(u_i, u_j) \in R_p \text{ if and only if } p(u_i) = p(u_j),$$

where $p(u_i)$ and $p(u_j)$ are the Boolean values of u_i and u_j in p , respectively.

- (3) Suppose that p is a categorical attribute in T . If u_i and u_j are classified in the same category by p , then $(u_i, u_j) \in R_p$; otherwise, $(u_i, u_j) \notin R_p$.

Subsequently, we obtained 11 relation decision systems induced by the 11 UCI datasets. Let $(U, C \cup D)$ be a relation decision system and $Red(C) = \{C_1, C_2, \dots, C_n\}$ be the set of all topological reducts (consistent topological reducts or general topological reducts). If $C_{min} \in Red(C)$ and $|C_{min}| = \min\{|C_i| | C_i \in Red(C)\}$, then C_{min} is called a minimal topological reduct. Generally, a relation decision system may have more than one minimal topological reduct. We used the efficient algorithm proposed by Liu and Feng [34] to identify the minimal topological reduct in the experiments. Then, for a relation decision system $(U, C \cup D)$ induced by one of the 11 UCI datasets, we acquired the corresponding reduced relation decision system $(U, C_{min} \cup D)$, denoted by $Red((U, C \cup D))$.

For convenience, we use “ $\mathbb{D}$ ” followed by the serial numbers in Table 2 to denote the relation decision systems induced by the corresponding experimental datasets. For example, $\mathbb{D}2$ denotes the relation decision system induced by the Wine dataset, for which the serial number is 2. Then the reduced $\mathbb{D}2$ can be denoted by $Red(\mathbb{D}2)$.

In our experiments, we used six classification algorithms and three clustering algorithms selected from machine learning to process the 11 relation decision systems induced by the 11 UCI datasets, and their reduced relation decision systems. In the experiments using classification algorithms, we focused on the training time (TT) and accuracy. In the experiments

Table 3
Comparison of the cardinality of the condition attribute sets.

Relation decision systems	$ C $	$ C_{min} $
$\mathbb{D}1$	18	5
$\mathbb{D}2$	13	5
$\mathbb{D}3$	9	4
$\mathbb{D}4$	14	8
$\mathbb{D}5$	18	6
$\mathbb{D}6$	22	6
$\mathbb{D}7$	12	8
$\mathbb{D}8$	50	7
$\mathbb{D}9$	44	7
$\mathbb{D}10$	12	6
$\mathbb{D}11$	147	7

Table 4
Performance of the SVM algorithm.

Relation decision systems	Raw		Reduced	
	TT (s)	Accuracy (%)	TT (s)	Accuracy (%)
$\mathbb{D}1$	0.00355	95.37	0.00191	91.48
$\mathbb{D}2$	0.05207	96.67	0.04338	93.89
$\mathbb{D}3$	0.00859	63.55	0.00768	58.96
$\mathbb{D}4$	0.00653	61.18	0.00622	62.06
$\mathbb{D}5$	0.00184	84.48	0.00139	81.00
$\mathbb{D}6$	0.10983	83.42	0.04013	85.03
$\mathbb{D}7$	0.01015	66.25	0.00828	65.08
$\mathbb{D}8$	0.00956	55.75	0.00849	51.75
$\mathbb{D}9$	0.00169	92.02	0.00152	92.02
$\mathbb{D}10$	0.00213	71.46	0.00198	71.46
$\mathbb{D}11$	0.03216	62.87	0.01384	66.71

using clustering algorithms, we paid attention to the running time (RT) and accuracy, evaluated by the adjusted Rand index (ARI) [63]. We used Python to write the code and ran the programs on a notebook computer with Windows 10, for which the hardware environment was an Intel(R) Core(TM) i5-8250U CPU @ 1.60GHz with 8 GB RAM.

The change in the condition attribute set of each relation decision system after reduction is shown in Table 3. The table shows that the dimensions of all the relation decision systems were reduced after reduction. For instance, the cardinality of the condition attribute set of $\mathbb{D}11$ was reduced from 147 to 7 after reduction.

6.1. Comparative experiments using classification algorithms

In the experiments, we used six machine learning classification algorithms, that is, the support vector machine (SVM), logistic regression (LR), decision tree (DT), k -nearest neighbors (KNN), linear discriminant analysis (LDA), and multilayer perceptron (MLP), to process the 11 relation decision systems and their corresponding reduced relation decision systems. Then we compared the performance of the six classification algorithms on each relation decision system and on the corresponding reduced relation decision system. We evaluated the performance using the TT and accuracy. Additionally, we defined accuracy as the mean value of the accuracies computed using ten-fold cross-validation.

Tables 4–9 present the experimental results. For the six classification algorithms, the TTs on all the reduced relation decision systems were less than those on the original relation decision systems, whereas the accuracies declined slightly in a little more than half of the cases. For example, Table 5 shows that the TT of the LR algorithm on $Red(\mathbb{D}2)$ was approximately 18.80% of that on $\mathbb{D}2$, and accuracy declined from 95.49% before reduction to 92.78% after reduction. It is important to note that the accuracy improved in more than one-third of the cases. For instance, as shown in Table 9, the accuracy of the MLP algorithm on $\mathbb{D}9$ increased from 78.01% before reduction to 92.02% after reduction. Furthermore, the TT of the MLP algorithm on $Red(\mathbb{D}9)$ was approximately 9.53% of that on $\mathbb{D}9$. In particular, some reduced relation decision systems enhanced computational efficiency greatly. Table 5 shows that the TT of the LR algorithm on $Red(\mathbb{D}11)$ was only approximately 0.59% of that on $\mathbb{D}11$. In a few cases, the reduced relation decision systems maintained accuracy. As shown in Table 6, the accuracy of the DT algorithm on $Red(\mathbb{D}1)$ was equal to that on $\mathbb{D}1$.

Table 4 shows the outcome of the experiments using the SVM algorithm. The TT of the SVM algorithm on each reduced relation decision system was less than that on the original relation decision system. Regarding accuracy, three of the 11 reduced relation decision systems led to an improvement, two were unchanged, and six led to a small fall.

Table 5 shows the performance of the LR algorithm. The TT of the LR algorithm on each reduced relation decision system was less than that on the original relation decision system. Four of the 11 reduced relation decision systems increased accuracy and seven reduced relation decision systems led to a modest fall in accuracy.

Table 5
Performance of the LR algorithm.

Relation decision systems	Raw		Reduced	
	TT (s)	Accuracy (%)	TT (s)	Accuracy (%)
D1	0.00153	93.15	0.00098	91.48
D2	0.02409	95.49	0.00453	92.78
D3	0.14127	63.10	0.12907	59.31
D4	0.07579	54.41	0.06652	49.12
D5	0.00480	82.48	0.00231	77.71
D6	0.00510	81.84	0.00098	85.45
D7	0.02095	64.47	0.02007	62.34
D8	0.02439	51.75	0.00822	47.25
D9	0.11104	82.84	0.03629	91.49
D10	0.00343	70.91	0.00217	71.46
D11	2.28135	71.77	0.01351	73.58

Table 6
Performance of the DT algorithm.

Relation decision systems	Raw		Reduced	
	TT (s)	Accuracy (%)	TT (s)	Accuracy (%)
D1	0.00095	91.48	0.00073	91.48
D2	0.00126	92.78	0.00082	90.00
D3	0.00182	58.44	0.00126	64.05
D4	0.00174	62.06	0.00141	61.18
D5	0.00105	77.00	0.00071	79.52
D6	0.00094	76.82	0.00075	79.39
D7	0.00240	58.94	0.00192	55.59
D8	0.00198	57.00	0.00111	55.50
D9	0.00241	84.01	0.00091	86.08
D10	0.00079	71.46	0.00067	71.46
D11	0.00511	73.79	0.00217	68.27

Table 7
Performance of the KNN algorithm.

Relation decision systems	Raw		Reduced	
	TT (s)	Accuracy (%)	TT (s)	Accuracy (%)
D1	0.00158	93.15	0.00095	91.30
D2	0.00107	74.80	0.00096	74.25
D3	0.00103	65.02	0.00096	65.52
D4	0.00166	65.29	0.00117	64.12
D5	0.00110	79.05	0.00096	79.67
D6	0.00136	74.68	0.00094	75.71
D7	0.00134	72.96	0.00102	69.95
D8	0.00240	50.75	0.00095	47.75
D9	0.00145	90.41	0.00091	92.02
D10	0.00101	71.46	0.00094	71.46
D11	0.01904	52.45	0.00142	60.76

Table 6 shows the outcome of the experiments using the DT algorithm. The TT of the DT algorithm on each reduced relation decision system was less than that on the original relation decision system. Four of the 11 reduced relation decision systems increased accuracy, two reduced relation decision systems maintained accuracy, and five reduced relation decision systems led to a small fall in accuracy.

Table 7 shows the performance of the KNN algorithm. The TT of the KNN algorithm on each reduced relation decision system was less than that on the original relation decision system. Five of the 11 reduced relation decision systems increased accuracy, *Red*(D10) maintained accuracy, and five reduced relation decision systems led to a modest fall in accuracy.

Table 8 shows the outcome of the experiments using the LDA algorithm. The TT of the LDA algorithm on each reduced relation decision system was less than that on the original relation decision system. Three of the 11 reduced relation decision systems increased accuracy and eight reduced relation decision systems led to a small fall in accuracy.

Table 9 shows the performance of the MLP algorithm. The TT of the MLP algorithm on each reduced relation decision system was less than that on the original relation decision system. Five of the 11 reduced relation decision systems increased accuracy, *Red*(D5) and *Red*(D10) maintained accuracy, and four reduced relation decision systems led to a minor decline in accuracy.

Table 8
Performance of the LDA algorithm.

Relation decision systems	Raw		Reduced	
	TT (s)	Accuracy (%)	TT (s)	Accuracy (%)
D1	0.00265	95.19	0.00104	91.48
D2	0.00223	97.78	0.00121	92.22
D3	0.00218	59.76	0.00161	53.18
D4	0.00627	67.65	0.00543	63.24
D5	0.00190	84.48	0.00112	80.38
D6	0.00198	80.76	0.00107	81.79
D7	0.00213	62.90	0.00159	59.30
D8	0.00740	67.00	0.00179	62.00
D9	0.00567	89.88	0.00130	92.02
D10	0.00148	70.35	0.00102	70.91
D11	0.02832	77.13	0.00200	75.74

Table 9
Performance of the MLP algorithm.

Relation decision systems	Raw		Reduced	
	TT (s)	Accuracy (%)	TT (s)	Accuracy (%)
D1	0.04008	95.56	0.01709	91.48
D2	0.12559	92.78	0.11777	92.22
D3	0.22473	57.94	0.20001	61.69
D4	0.74746	75.59	0.72301	71.18
D5	0.22175	79.05	0.09970	79.05
D6	0.28267	81.29	0.15238	82.39
D7	0.17551	68.13	0.15720	64.79
D8	0.27404	52.25	0.23180	54.25
D9	0.32060	78.01	0.03055	92.02
D10	0.04650	71.46	0.01988	71.46
D11	1.49805	51.64	0.60004	71.64

Table 10
Performance of the K-means algorithm.

Relation decision systems	Raw		Reduced	
	RT (s)	ARI	RT (s)	ARI
D1	0.00768	0.0110	0.00579	-0.0007
D2	0.00597	0.3711	0.00442	0.3711
D3	0.00674	0.2546	0.00551	0.2805
D4	0.01972	0.2790	0.01855	0.3294
D5	0.02053	0.1230	0.01838	0.1254
D6	0.00603	0.2135	0.00427	0.1295
D7	0.07426	0.0434	0.07004	0.0590
D8	0.02366	0.0327	0.01796	0.0268
D9	0.00544	-0.0839	0.00436	-0.0832
D10	0.00477	-0.0041	0.00395	-0.0026
D11	0.01387	0.0382	0.00955	0.1211

6.2. Comparative experiments using clustering algorithms

We used three clustering algorithms, that is, the K-means, agglomerative nesting (AGNES), and spectral clustering (SC), to cluster the 11 original relation decision systems and the corresponding reduced relation decision systems. Moreover, we evaluated the performance of each clustering algorithm using the RT and ARI values as the evaluation indices. The ARI value is in the interval $[-1, 1]$; the higher the ARI value, the better.

Tables 10–12 show the experimental results. All reduced relation decision systems enabled the three clustering algorithms to reduce the RT; however, they decreased the ARI values slightly in a minority of cases. For instance, Table 11 shows that the RT of the AGNES algorithm on $Red(D3)$ was approximately 44.52% of that on D3, whereas the ARI value declined from 0.0206 before reduction to 0.0149 after reduction. In most instances, the reduced relation decision systems increased the ARI values. For example, the ARI value increased from -0.0446 to 0.0047 when we used the SC algorithm to cluster $Red(D1)$, which is shown in Table 12. Furthermore, the RT of the SC algorithm on $Red(D1)$ was approximately 42.74% of that on D1. In rare cases, the reduced relation decision systems kept the ARI values unchanged. For example, $Red(D2)$ maintained the ARI value when we used the K-means algorithm to cluster it, which is shown in Table 10.

Table 11
Performance of the AGNES algorithm.

Relation decision systems	Raw		Reduced	
	RT (s)	ARI	RT (s)	ARI
D1	0.00808	-0.0590	0.00631	-0.0005
D2	0.00173	0.3684	0.00092	0.3627
D3	0.00155	0.0206	0.00069	0.0149
D4	0.00390	0.2729	0.00261	0.3274
D5	0.00157	0.1071	0.00075	0.2209
D6	0.00214	0.0058	0.00111	0.1162
D7	0.00360	-0.0814	0.00240	0.0526
D8	0.00716	0.0454	0.00397	0.0247
D9	0.00248	-0.0782	0.00118	-0.0753
D10	0.00171	-0.0072	0.00111	0.0014
D11	0.01898	-0.0045	0.00604	0.0749

Table 12
Performance of the SC algorithm.

Relation decision systems	Raw		Reduced	
	RT (s)	ARI	RT (s)	ARI
D1	0.12565	-0.0446	0.05370	0.0047
D2	0.00934	0.3874	0.00886	0.3874
D3	0.05008	0.1669	0.03273	0.1667
D4	0.78925	0.3084	0.53854	0.3268
D5	0.01280	0.1277	0.00858	0.2327
D6	0.00981	-0.0523	0.00835	0.1292
D7	0.07902	0.0851	0.06790	0.1172
D8	0.03119	0.0326	0.02070	0.0256
D9	0.02626	-0.0040	0.02067	0.0186
D10	0.03057	0.0006	0.02373	-0.0062
D11	0.01989	0.0020	0.01247	0.0020

Table 10 shows the outcome of the comparison using the *K*-means algorithm. The RT of the *K*-means algorithm on each reduced relation decision system was less than that on the original relation decision system. Regarding the ARI values, seven of the 11 reduced relation decision systems enhanced the ARI values, *Red*(D2) kept the ARI value unchanged, and three reduced relation decision systems led to a small fall in the ARI values.

Table 11 shows the results of the comparative experiments using the AGNES algorithm. The RT of the AGNES algorithm on each reduced relation decision system was less than that on the original relation decision system. Moreover, eight of the 11 reduced relation decision systems increased the ARI values and three reduced relation decision systems led to a minor decline in the ARI values.

Table 12 shows the performance of the SC algorithm. The RT of the SC algorithm on each reduced relation decision system was less than that on the original relation decision system. Additionally, six of the 11 reduced relation decision systems enhanced the ARI values, *Red*(D2) and *Red*(D11) kept the ARI values unchanged, and three reduced relation decision systems led to a modest fall in the ARI values.

7. Discussion

Decision-making problems are widespread in the real world. Many of them depend upon uncertain data. Some mathematical theories are used to solve decision-making problems of this type. For example, Akram et al. [64] introduced a novel hybrid model called the interval-valued *m*-polar fuzzy soft set and developed an algorithm to solve decision-making problems that have data in interval-valued *m*-polar fuzzy soft form. The theory of rough sets is a mathematical theory that can be used to address various types of uncertainties in real-world decision-making problems [65]. Most decision-making problems can be formally represented by decision systems, such as decision tables, relation decision systems, and bipolar fuzzy relation decision systems [37]. Compared with information systems, incomplete information systems, and relation systems, which do not have decision attribute sets, decision systems are more suitable for real-world decision-making problems. They have been widely used in fields in which rough set theory has made many achievements, such as expert systems, decision support systems, knowledge acquisition, and supervised learning.

There is a close relationship between topologies and rough sets. Pawlak [1] not only proposed rough sets, but also generated a special type of topology, which is called Pawlak topology [66]. Undoubtedly, investigating the theory of rough sets from the topological viewpoint has deep theoretical and practical significance. Some authors [32,47,56–58] have considered the problem of attribute reduction in rough sets using topological methods and proposed various definitions of topological reduction. However, the concept of topological reduction has not been introduced to decision systems in rough sets.

Consequently, to establish the theory of topological reduction of decision systems in rough sets, we innovatively proposed the concepts of consistent topological reduction and general topological reduction of relation decision systems. Moreover, we obtained the corresponding reduction algorithms based on discernibility matrices to compute all the reducts. In fact, the consistent topological reduction is a special case of general topological reduction and general topological reduction is the natural generalization of consistent topological reduction. Therefore, Algorithm 4.1 can be viewed as a generalization of Algorithm 3.1. Thus, Algorithm 4.1 has a wider range of applications than Algorithm 3.1. Using Example 5.1, we showed that Algorithms 3.1 and 4.1 cannot be unified by the GRA, which greatly unifies earlier attribute reduction algorithms for usual decision tables. However, an open problem is what is the relationship between general topological reducts (not consistent topological reducts) and the reducts identified by the GRA, which requires exploration in future work.

The results of the experiments on real-world data confirmed that the present study is meaningful. However, the proposed reduction algorithms are time-consuming. Therefore, we should explore how to reduce the cost of computation. Moreover, in the future, we should consider how to find the optimal topological reducts straightforwardly for practical problems.

8. Conclusions

Attribute reduction is an essential and effective data processing technology, which has gained a great deal of attention. However, the concept of topological reduction has not been introduced to decision systems in rough sets. To fill the gap, we investigated the problem of topological reduction in relation decision systems in this study. We proposed the notions of consistent topological reduction and general topological reduction in relation decision systems. Moreover, we developed corresponding reduction algorithms based on discernibility matrices to identify all reduction sets. We discussed the relationship between consistent topological reducts and reducts identified by the GRA. We demonstrated that our reduction algorithms cannot be unified by the GRA. To verify our theoretical results, we conducted comparative experiments on 11 UCI datasets. The experimental results showed that the proposed reduction algorithms were effective and practicable. Moreover, it should be indicated that this study provides a new insight into topological reduction.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have shared the link to my data at the Attach File step

[Climate Model Simulation Crashes \(Reference Data\)](#) (UCI Machine Learning Repository)
[Wine \(Reference Data\)](#) (UCI Machine Learning Repository)
[Glass Identification \(Reference Data\)](#) (UCI Machine Learning Repository)
[Leaf \(Reference Data\)](#) (UCI Machine Learning Repository)
[Lymphography \(Reference Data\)](#) (UCI Machine Learning Repository)
[Parkinsons \(Reference Data\)](#) (UCI Machine Learning Repository)
[Speaker Accent Recognition \(Reference Data\)](#) (UCI Machine Learning Repository)
[Turkish Music Emotion Dataset \(Reference Data\)](#) (UCI Machine Learning Repository)
[SPECTF Heart \(Reference Data\)](#) (UCI Machine Learning Repository)
[Planning Relax \(Reference Data\)](#) (UCI Machine Learning Repository)
[Urban Land Cover \(Reference Data\)](#) (UCI Machine Learning Repository)

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